Test Design Techniques: Decision Table Testing

Decision Table Testing

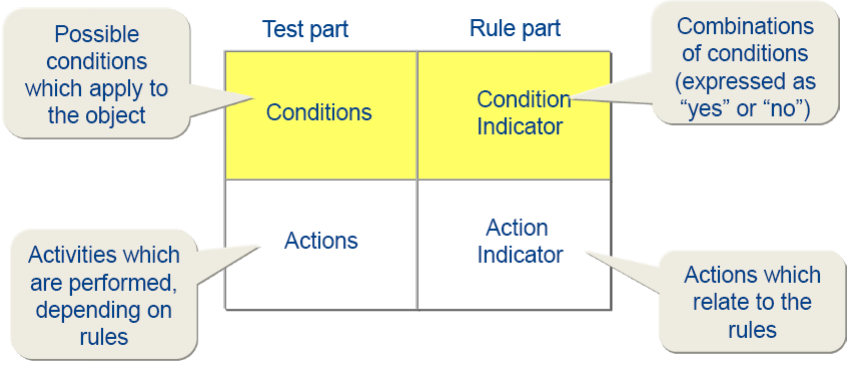
- Applicable for system requirements or specifications, which contain logical conditions and complex business rules that a system is to implement.
- Check specifications and define input conditions and actions of the system and state them in a way, that they can be either "true" or "false"
- Decision table contains combinations of "true" and "false" for all input conditions and the resulting actions.

Decision Table (cont.)

- Each column of the table corresponds to a rule in the business process, which defines a unique combination of conditions, which again results in the execution of the action, which is associated with this rule.
- For decision table testing usually used standard-coverage:
  - At least one test case per column (includes the coverage of all combinations of triggering conditions).

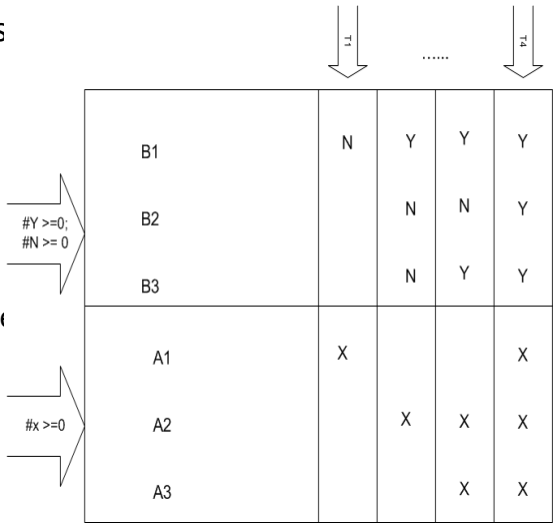
Decision Table Testing (cont.)

- The four quadrants of the decision table:



Decision Table

- Each column responds to a test case (rule)
- Coverage Criteria:
  - All conditions at least once N and Y
  - All actions at least onc x.
  - All condition-combinations.



A Simple Example

- Scenario: A marketing company wishes to construct a decision table to decide how to treat clients according to three characteristics:
  - Gender,
  - City Dweller, and
  - age group: A (under 30), B (between 30 and 60), C (over 60).
- The company has four products (W, X, Y and Z) to test market.
  - Product W will appeal to male city dwellers.
  - Product X will appeal to young males.
  - Product Y will appeal to Female middle aged shoppers who do not live in cities.
  - Product Z will appeal to all but older males.

1. Identify Conditions & Values

- The three data attributes tested by the conditions in this problem are
  - gender, with values M and F;
  - city dweller, with value Y and N; and
  - age group, with values A, B, and C
- as stated in the problem.

2. Compute Maximum Number of Rules

- The maximum number of rules is 2 x 2 x 3 = 12



– gender, with values M and F;
– city dweller, with value Y and N; and
– age group, with values A, B, and C

3. Identify Possible Actions

- The four actions are:
  - market product W,
  - market product X,
  - market product Y,
  - market product Z.

- Product W will appeal to male city dwellers.
- Product X will appeal to young males.
- Product Y will appeal to Female middle aged shoppers who do not live in cities.
- Product Z will appeal to all but older males.

4. Enter All Possible Conditions

RULES												
	1	2	3	4	5	6	7	8	9	10	11	12
Sex	m	f	m	f	m	f	m	f	m	f	m	f
City	y	y	n	n	y	y	n	n	y	Y	n	n
Age	a	a	a	a	b	b	b	b	c	c	c	c

– gender, with values M and F;
– city dweller, with value Y and N; and
– age group, with values A, B, and C

5. Define Actions for each Rule

Actions												
Market	1	2	3	4	5	6	7	8	9	10	11	12
W	X				X				X			
X	X		X									
Y								X				
Z	X	X	X	X	X	X	X	X		X		X

- Product W will appeal to male city dwellers.
- Product X will appeal to young males.
- Product Y will appeal to Female middle aged shoppers who do not live in cities.
- Product Z will appeal to all but older males.

5. Full table

RULES												
	1	2	3	4	5	6	7	8	9	10	11	12
Sex	m	f	m	f	m	f	m	f	m	f	m	f
City	y	y	n	n	y	y	n	n	y	Y	n	n
Age	a	a	a	a	b	b	b	b	c	c	c	c
Actions												
Market	1	2	3	4	5	6	7	8	9	10	11	12
W	X				X				X			
X	X		X									
Y								X				
Z	X	X	X	X	X	X	X	X		X		X

6. Verify the Policy

- Let us assume that the client agreed with our decision table.

7. Simplify the Table

- There appear to be no impossible rules.
- Note that rules 2, 4, 6, 7, 10, 12 have the same action pattern.
- Rules 2, 6 and 10 have
  - two of the three condition values (gender and city dweller) identical and
  - all three of the values of the non- identical value (age) are covered,
- so they can be condensed into a single column 2.
- The rules 4 and 12 have identical action pattern, but they cannot be combined because the indifferent attribute "Age" does not have all its values covered in these two columns.
- Age group B is missing.

8. Simplify the Table

RULES												
	1	2	3	4	5	6	7	8	9	10	11	12
Sex	m	f	m	f	m	f	m	f	m	f	m	f
City	y	y	n	n	y	y	n	n	y	Y	n	n
Age	a	**a**	a	a	b	**b**	b	b	c	**c**	c	c
Actions												
Market	1	2	3	4	5	6	7	8	9	10	11	12
W	X				X				X			
X	X		X									
Y								X				
Z	X	X	X	X	X	X	X	X		X		X

- two of the three condition values (gender and city dweller) identical and
- all three of the values of the non-identical value (age) are covered,
- so they can be condensed into a single column 2

9. The revised table is as follows:

RULES										
	1	2	3	4	5	6	7	8	9	10
Sex	M	F	M	F	M	M	F	M	M	F
City	Y	Y	N	N	Y	N	N	Y	N	N
Age	A	-	A	A	B	B	B	C	C	C
Actions										
Market	1	2	3	4	5	6	7	8	9	10
W					X			X		
X	X		X							
Y							X			
Z	X	X	X	X	X	X	X			X

10. Test Cases

Test Case ID	Condition 1	Condition 2	Condition 3	Expected Output	Actual Output	Pass/Fail
TC1	M	Dweller	29	Market X & Z		
:						
TC10						

Decision Table Methodology

1. Identify Conditions & Values	Find the data attribute each condition tests and all of the attribute's values.
2. Compute Max Number of Rules	Multiply the number of values for each condition data attribute by each other.
3. Identify Possible Actions	Determine each independent action to be taken for the decision or policy.
4. Enter All Possible Rules	Fill in the values of the condition data attributes in each numbered rule column.
5. Define Actions for each Rule	For each rule, mark the appropriate actions with an X in the decision table.
6. Verify the Policy	Review completed decision table with end-users.
7. Simplify the Table	Eliminate and/or consolidate rules to reduce the number of columns.